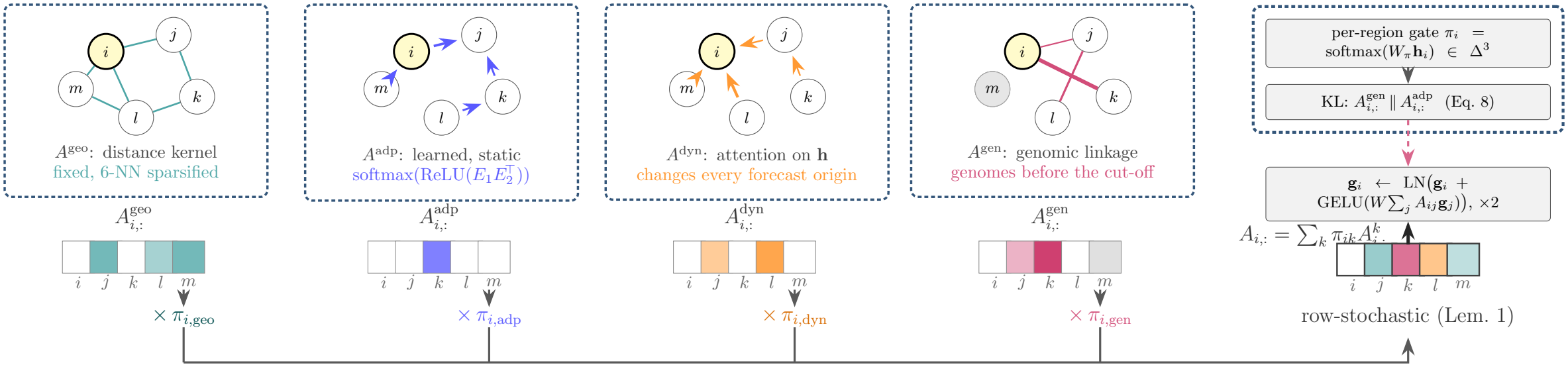

A^{geo}: distance kernel
fixed, 6-NN sparsified
A^{adp}: learned, static
$\text{softmax}(\text{ReLU}(E_1 E_2^\top))$
A^{dyn}: attention on $\mathbf{h}$
changes every forecast origin
A^{gen}: genomic linkage
genomes before the cut-off
$A^{\text{geo}}_{i,:}$
$A^{\text{adp}}_{i,:}$
$A^{\text{dyn}}_{i,:}$
$A^{\text{gen}}_{i,:}$
i j k l m
$\times\, \pi_{i,\text{geo}}$
$\times\, \pi_{i,\text{adp}}$
$\times\, \pi_{i,\text{dyn}}$
$\times\, \pi_{i,\text{gen}}$
per-region gate π_i = $\text{softmax}(W_\pi \mathbf{h}_i) \in \Delta^3$
KL: $A^{\text{gen}}_{i,:} \,\|\, A^{\text{adp}}_{i,:}$ (Eq. 8)
$\mathbf{g}_i \leftarrow \text{LN}\big(\mathbf{g}_i + \text{GELU}(W \sum_j A_{ij} \mathbf{g}_j)\big)$, $\times 2$
$A_{i,:} = \sum_k \pi_{ik} A^k_{i,:}$
row-stochastic (Lem. 1)